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Newton's 3 Laws of Motion

SS ILP Physics (Intermediate), Week 2 Class 3

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

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Outline

- Newton’s three Laws of Motion
- Procedures to solve dynamics problems with Newton’s three laws

Newton’s 1st Law of Motion

- An object remains its state of rest or uniform motion at a constant velocity unless it is acted upon a non-zero resultant force.

$$\sum \vec{\mathbf{F}}_i = \overrightarrow{\mathbf{O}} \rightarrow \vec{\mathbf{v}} = \text{constant}$$

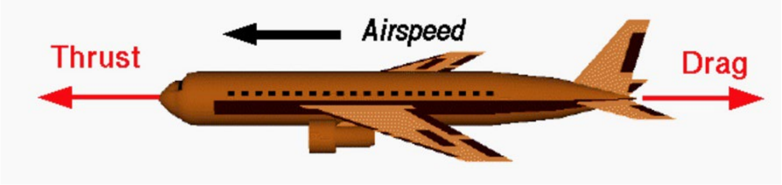
https://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law

First law

Some examples are:

https://mammothmemory.net/physics/newtons-laws-of-motion/newtons-first-law--examples/newtons-first-law-examples.html

When a car brakes quickly, the passenger will be thrown forward because inertia (the tendency to remain unchanged) tries to keep the passenger moving.



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An aircraft in flight is a particularly good example of the first law of motion. There are four major forces acting on an aircraft; lift, weight, thrust, and drag. At a constant altitude, we can neglect the lift and weight.

A cruising aircraft flies at a constant airspeed and the thrust exactly balances the drag of the aircraft. There is no net force on the airplane, and it travels at a constant velocity in a straight line.

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Newton's Second Law

- The acceleration is proportional to the resultant force, in the same direction and inversely proportional to the mass,

$$\boxed{\sum \vec{\mathbf{F}}_i = \frac{d(m\vec{\mathbf{v}})}{dt} = m\vec{\mathbf{a}}}$$

*Sum of all the forces acting on that object

https://www.physicsclassroom.com/class/newtlaws/Lesson-3/Newton-s-Second-Law

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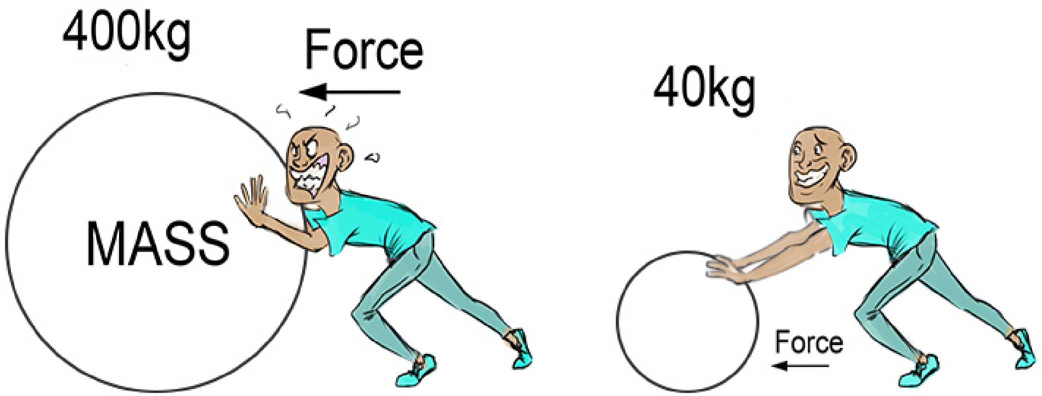
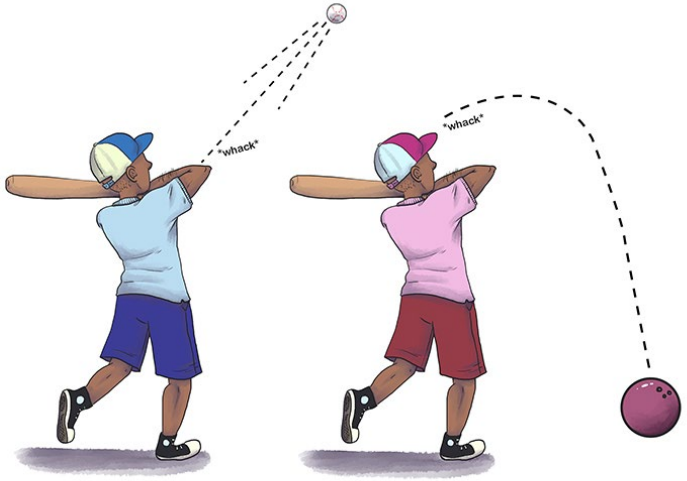
Some examples are:



https://www.onespecialscienceteacher.com/wp-content/uploads/2021/01/newtons-second-law-pushing-ball-version-2-.94b7355.jpg



https://clarkscience8.weebly.com/force-vs-mass-fma.html text

If you think of acceleration as movement, then:
"The greater the mass of the object, the more force needed to make it accelerate"
can be read as:
"The greater the mass of the object the more force is needed to make it move."

Because the mass of each ball is different, each ball will travel a different distance and at a different speed when it is hit with the same force.

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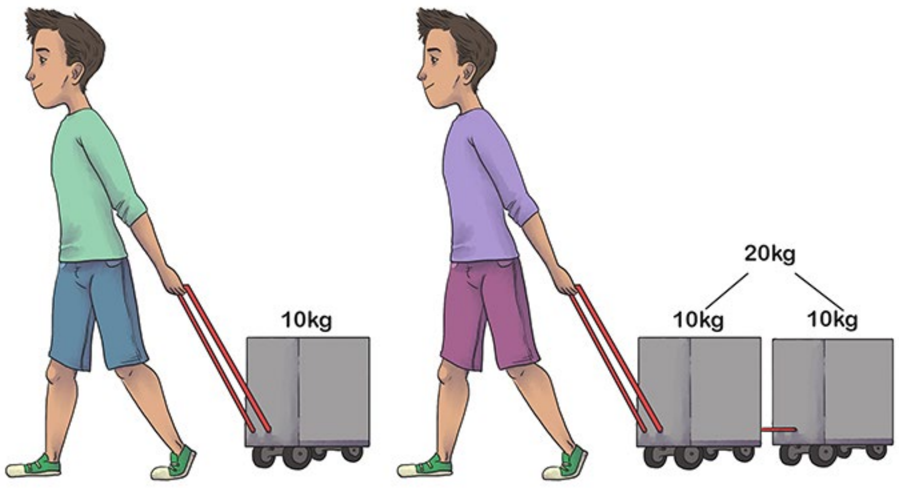
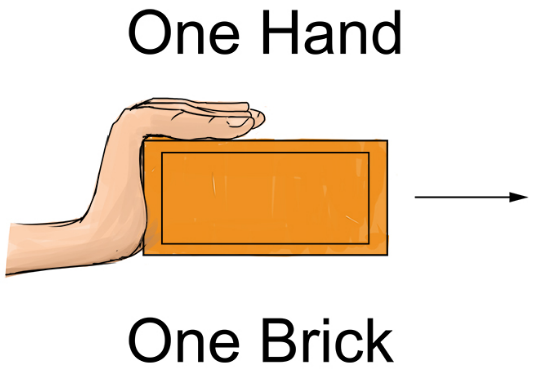


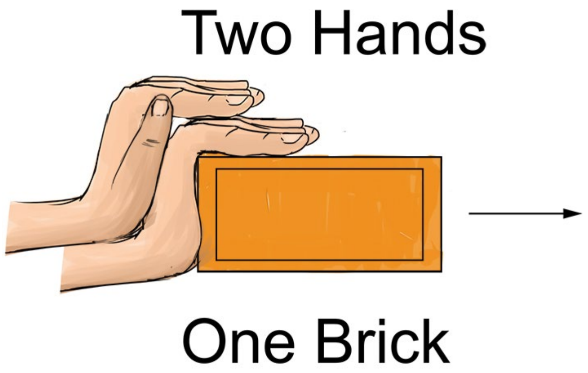
It will take twice the amount of force to accelerate the wagon with 20kg as the wagon with 10kg.

But to make an object accelerate or move you must apply a force.

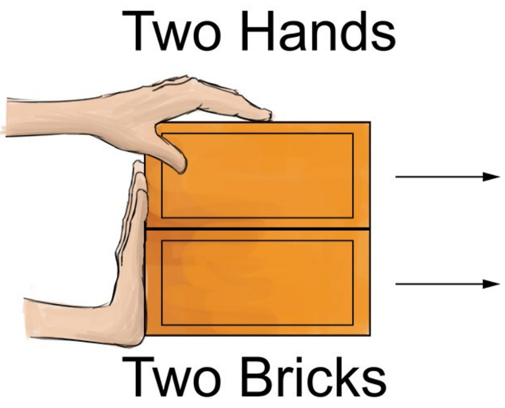
Click to https://mammothmemory.net/physics/newtons-laws-of-motion/newtons-second-law--examples/newtons-second-law-examples.htmladd text



Force of hand accelerates the brick



Twice as much force produces twice as much acceleration.



Twice the force on twice the mass gives the same acceleration.

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Example

Moe, Larry, and Curly push on a 752-kg boat that floats next to a dock. They each exert an 80.5N force parallel to the dock.

(a) What is the acceleration of the boat if they all push in the same direction?

(b) What is the magnitude and direction of the boat's acceleration if Larry and Curly push in the opposite direction to Moe's push?

J. S. Walker, Physics, Example 5-1, p115

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Example: Solution

Moe, Larry, and Curly push on a 752-kg boat that floats next to a dock. They each exert an 80.5N force parallel to the dock.

(a) What is the acceleration of the boat if they all push in the same direction?

$$\sum F = ma$$

$$F_M\hat{\imath} + F_L\hat{\imath} + F_C\hat{\imath} = ma\,\hat{\imath}$$

$$F_M + F_L + F_C = ma \quad \rightarrow \quad 80.5 \times 3 = 752\,a \quad \rightarrow \quad a = 0.321\,m/s^2$$

J. S. Walker, Physics, Example 5-1, p115

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Example: Solution

Moe, Larry, and Curly push on a 752-kg boat that floats next to a dock. They each exert an 80.5N force parallel to the dock.

(b) What is the magnitude and direction of the boat's acceleration if Larry and Curly push in the opposite direction to Moe's push?

$$\sum F = ma$$

$$F_M(-\hat{\imath}) + F_L\hat{\imath} + F_C\hat{\imath} = ma\,\hat{\imath}$$

$$-F_M + F_L + F_C = ma \quad \rightarrow \quad 80.5 = 752\,a \quad \rightarrow \quad a = 0.107\,m/s^2$$

J. S. Walker, Physics, Example 5-1, p115

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Newton's Third Law

- To every action, there is always an opposite reaction of the same magnitude,

$$\vec{\boldsymbol{F}}_{2,1} = -\vec{\boldsymbol{F}}_{1,2}$$

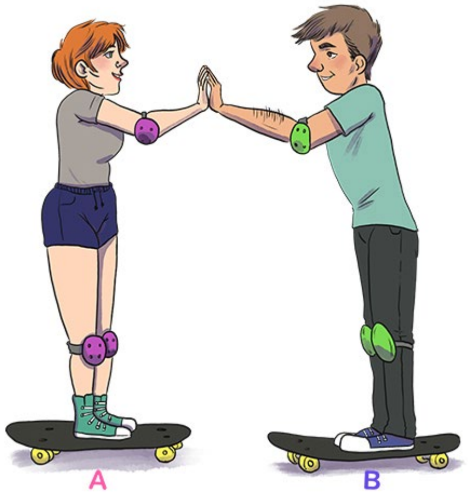
force on object 2 due to object 1

force on object 1 due to object 2

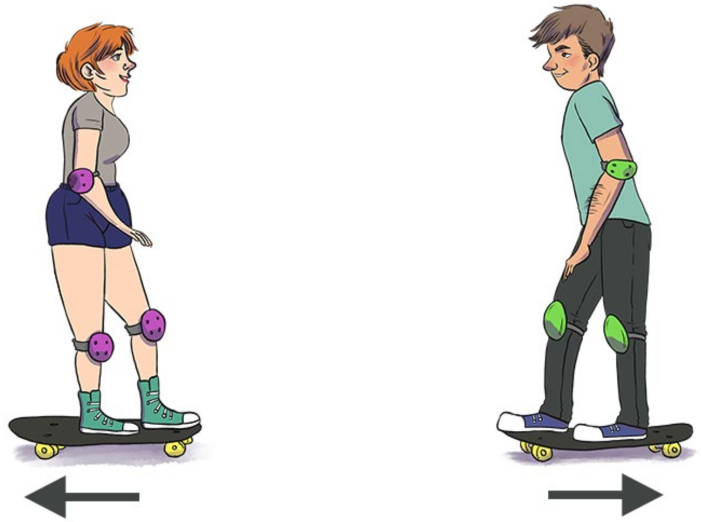
https://www.physicsclassroom.com/class/newtlaws/Lesson-4/Newton-s-Third-Law

Third law

Some examples are:



Skater A pushes against skater B.



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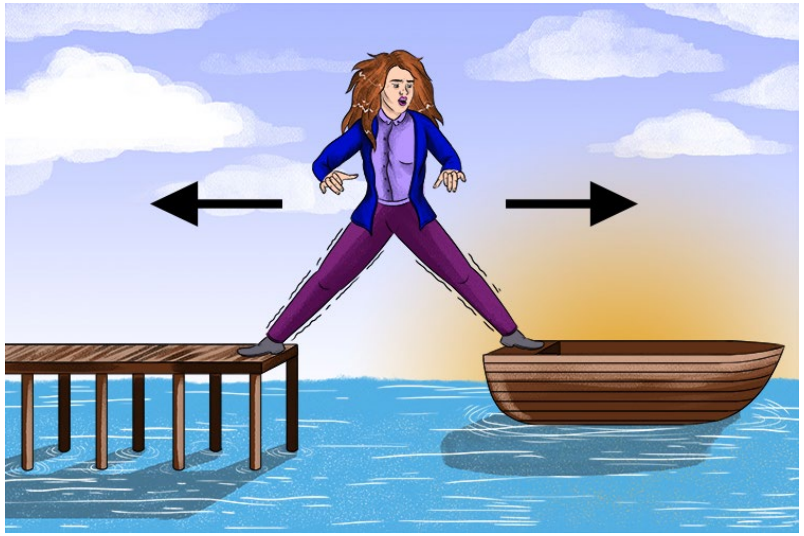
Skater B will accelerate to the right according to $F = ma$

Skater A will accelerate to the left because there is an equal and opposite force.

A woman on a boat tries to step off the boat on to a pier

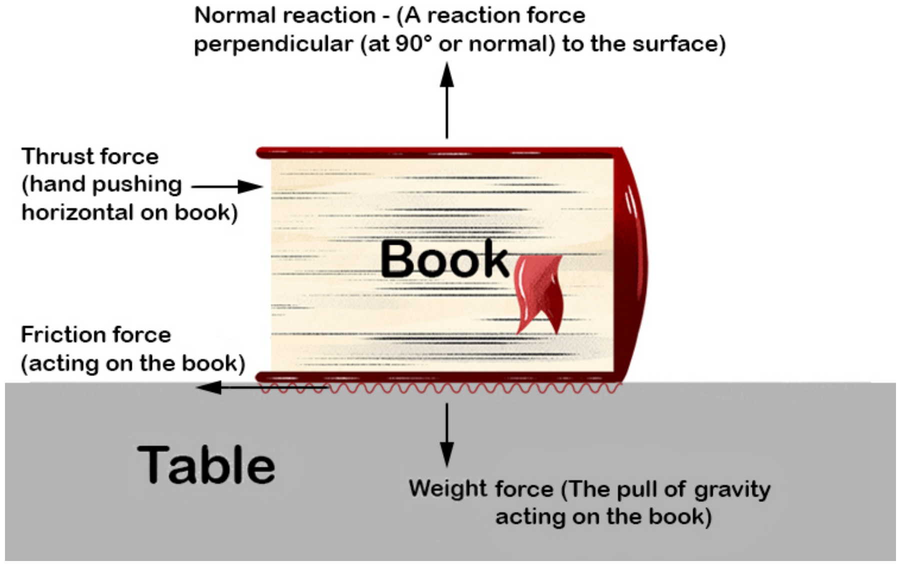


For every action, *there is an equal and opposite reaction*. Pushing your body forward will have an equal reaction backwards on the boat.

NOTE: Forces always come in pairs: that is why Newton's third law is sometimes referred to as his law of pairs.

https://mammothmemory.net/physics/newtons-laws-of-motion/newtons-third-law--examples/newtons-third-law-examples.html

A book being pushed across a table.



This book being pushed along shows how forces come in pairs.

The weight of the book exerts a force downward and the table needs to exert an equal force upward or the table will collapse.

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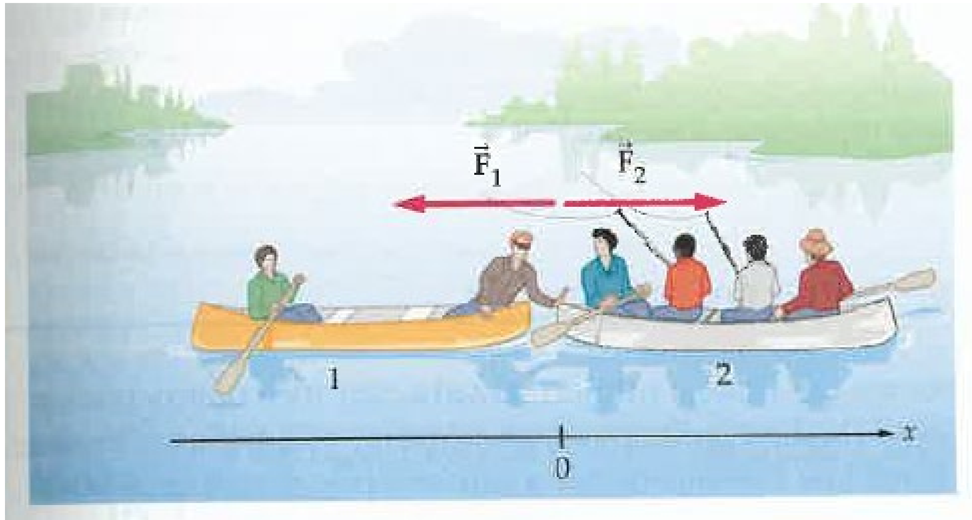
Example

Two groups of canoeists meet in the middle of a lake. After a brief visit, a person in canoe 1 pushes on canoe 2 with a force of 46 N to separate the canoes. If the mass of canoe 1 and its occupants is $m_1 = 150kg$, and the mass of canoe 2 and its occupants is $m_2 = 250kg$, find the acceleration the push gives to each canoe.



Physical picture

J. S. Walker, Physics, Example 5-3, p119

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Example: Solution

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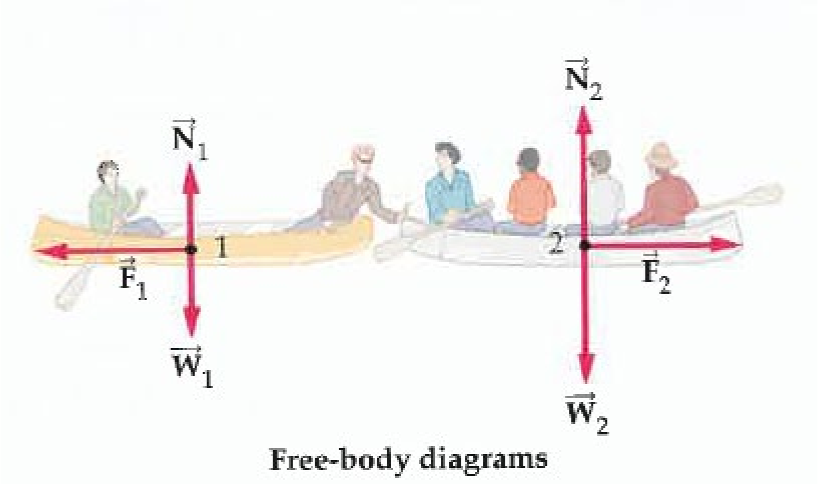

$$\sum F = ma$$

$$F_1(-\hat{\imath}) = m_1 a_1 \hat{\imath} \quad \rightarrow \quad a_1 = -0.31\, m/s^2$$

$$F_2 \hat{\imath} = m_2 a_2 \hat{\imath} \quad \rightarrow \quad a_2 = 0.18\, m/s^2$$

J. S. Walker, Physics, Example 5-3, p119

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Newton's three Laws of Motion

- An object remains its state of rest or uniform motion at a constant velocity unless it is acted upon a non-zero resultant force.

$$\sum \vec{\mathbf{F}}_i = \overrightarrow{\mathbf{0}} \rightarrow \vec{\mathbf{v}} = \text{constant}$$

- The acceleration is proportional to the resultant force, in the same direction and inversely proportional to the mass,

$$\sum \vec{\mathbf{F}}_i = \frac{d(m\vec{\mathbf{v}})}{dt} = m\vec{\mathbf{a}}$$

- To every action, there is always an opposite reaction of the same magnitude.

$$\overrightarrow{\boldsymbol{F}}_{2,1} = -\overrightarrow{\boldsymbol{F}}_{1,2}$$

force on object 2 due to object 1 force on object 1 due to object 2

https://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law
https://www.physicsclassroom.com/class/newtlaws/Lesson-3/Newton-s-Second-Law
https://www.physicsclassroom.com/class/newtlaws/Lesson-4/Newton-s-Third-Law

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Solving dynamics problems with Newton's three laws

I. Identify the object(s) of interest.

II. Draw free body diagram(s) of the object(s).

III. Choose a proper coordinate system and determine the components of the forces and **acceleration** along coordinate axes.

IV. Apply Newton's 2nd law of motion to write equations with unknowns,

$$\sum_{i}^{N} \vec{F}_i = m\vec{a}$$

V. In addition to Newton's 2nd law, identify any other equations you might need for solving unknowns.

H. D. Young and R. A. Freedman, University Physics (14th edition), p160.

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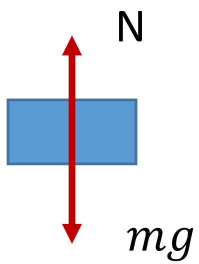
Example

A woman of mass *m* is standing on a weighing scale in a lift. If the lift is accelerating downwards at 0.1g, determine the effective weight of the woman shown on the scale.

Example: Solution

- Must draw free body diagram



- Taking upwards positive, and using $\sum \vec{\mathbf{F}}_i = m\vec{\mathbf{a}}$

$$-mg + N = m\,(-0.1g)$$

Hence, $N = 0.9\,mg$

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Case problem 1

A woman of mass *m* is standing on a weighing scale in a lift. If the lift is accelerating upwards at 0.1g, determine the effective weight of the woman shown on the scale.

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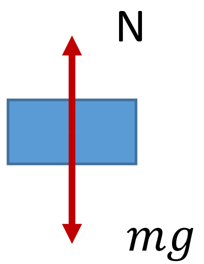
Case problem 1 - Solution

- Must draw free body diagram



- Taking upwards positive, and using $\sum \vec{\mathbf{F}}_i = m\vec{\mathbf{a}}$

$$N - mg = m\,(0.1g)$$

Hence, $N = 1.1\,mg$

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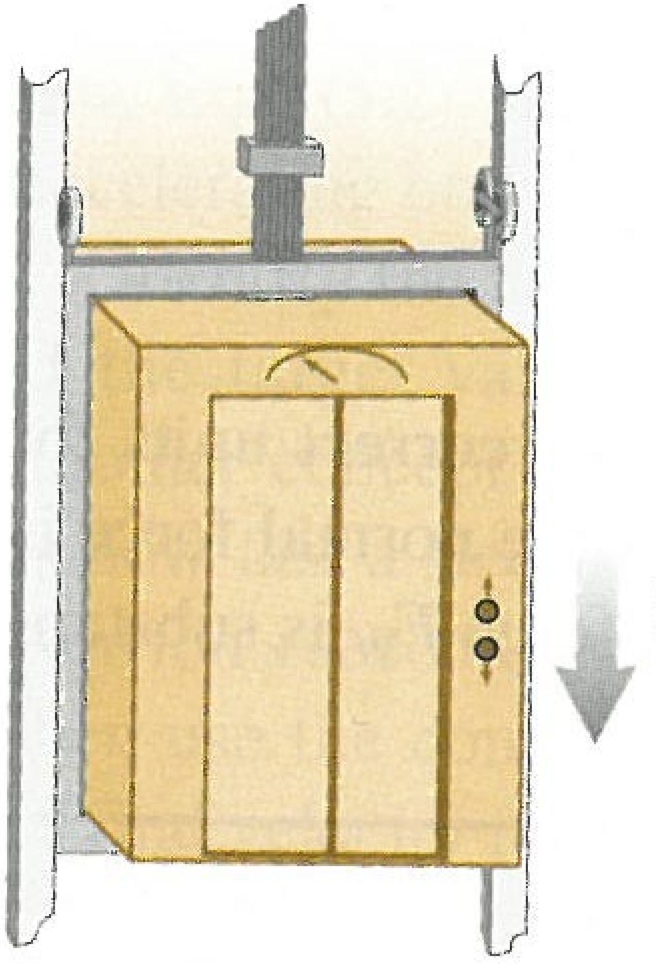
Case problem 2

An elevator and its load have a combined mass of 800 kg. The elevator is initially moving downward at 10.0 m/s; it slows to a stop with constant acceleration in a distance of 25.0 m. What is the tension T in the supporting cable while the elevator is being brought to rest? Take $g = 9.8\, m/s^2$.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.8, p162

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Case problem 2 - Solution

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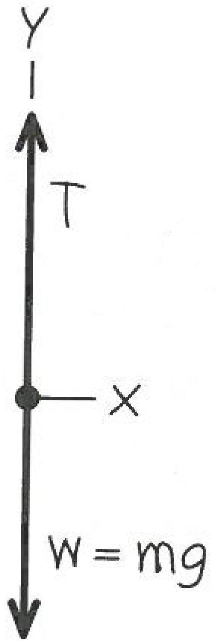


$$v_y^2 - v_{y0}^2 = 2a_y(y - y_0)$$

$$a_y = \frac{v_y^2 - v_{y0}^2}{2(y - y_0)} = 2\, m/s^2$$

$$\sum F_y = ma_y$$

$$T \ - mg = ma_y$$

$$T = mg + ma_y$$

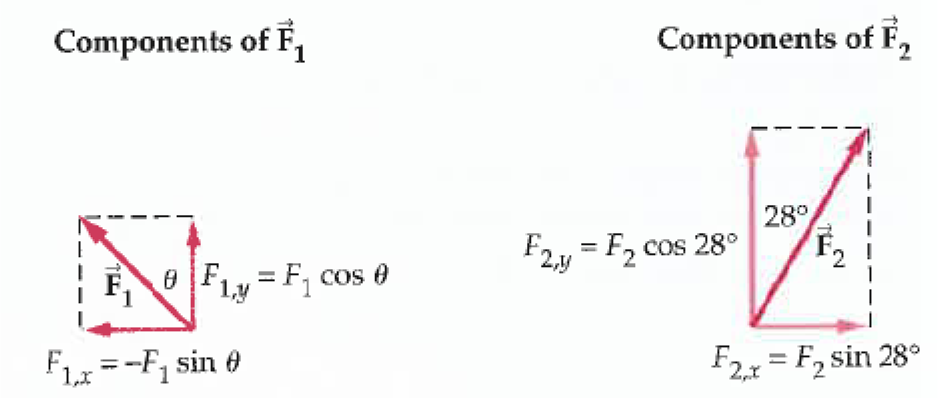
$$T = 9440\ N$$

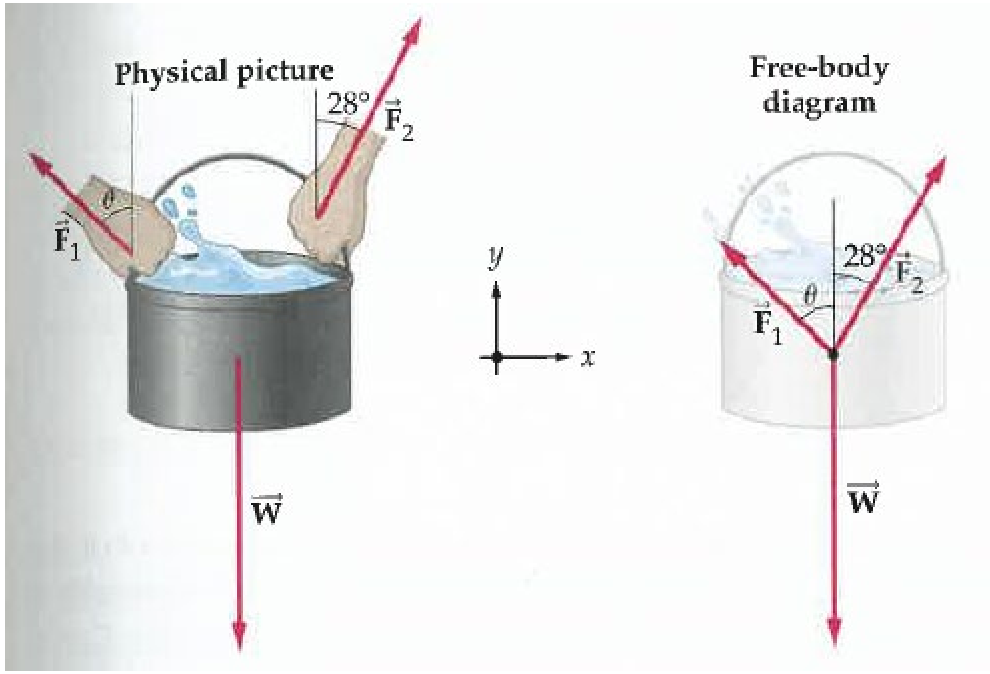
H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.8, p162

Forces in two dimensions

- When the forces are different in magnitude and directions, treat each coordinate direction independently of the other.

$$\sum F = ma \rightarrow \begin{array}{l} \sum F_x = ma_x \\ \sum F_y = ma_y \end{array}$$





J. S. Walker, Physics, p121-123

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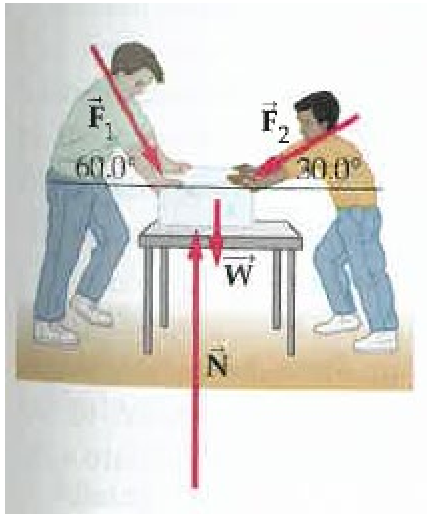
Example

A 6.0-kg block of ice is acted on by two forces, $\vec{F}_1$ and $\vec{F}_2$, as shown in the diagram. If the magnitudes of the forces are $F_1 = 13\text{ N}$ and $F_2 = 11\text{ N}$, find

(a) the acceleration of the ice and

(b) the normal force exerted on it by the table. Take $g = 9.8\ m/s^2$.



Physical picture

J. S. Walker, Physics, Example 5-8, p129

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Example: Solution

A 6.0-kg block of ice is acted on by two forces, $\vec{F}_1$ and $\vec{F}_2$, as shown in the diagram. If the magnitudes of the forces are $F_1 = 13$ N and $F_2 = 11$ N, find

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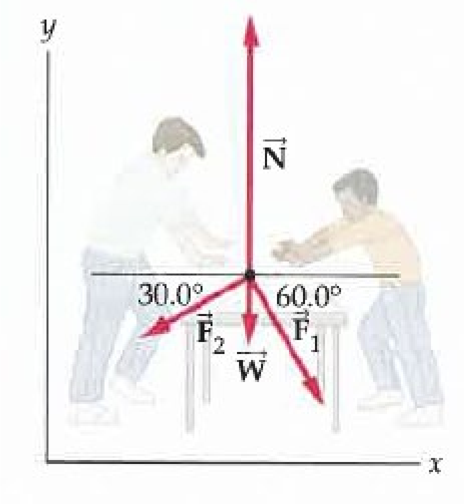

Free-body diagram

$$\sum F_x = ma_x \rightarrow \quad F_{1,x} - F_{2,x} = ma_x \rightarrow \quad a_x = -0.5\, m/s^2$$

$$\sum F_y = ma_y \qquad\qquad a_y = 0$$

J. S. Walker, Physics, Example 5-8, p129

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Example: Solution

A 6.0-kg block of ice is acted on by two forces, $\vec{F}_1$ and $\vec{F}_2$, as shown in the diagram. If the magnitudes of the forces are $F_1 = 13$ N and $F_2 = 11$ N, find

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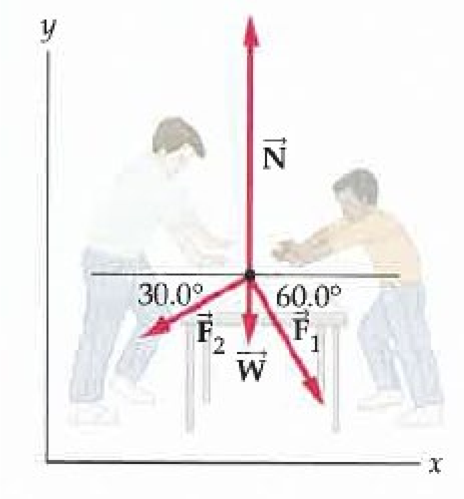

Free-body diagram

$$\sum F_x = ma_x$$

$$\sum F_y = ma_y \rightarrow \quad N - mg - F_{1,y} - F_{2,y} = 0 \ \rightarrow N = 75\ \mathrm{N}$$

$$\vec{N} = 75.6\ \mathrm{N}\ \hat{\jmath}$$

J. S. Walker, Physics, Example 5-8, p129

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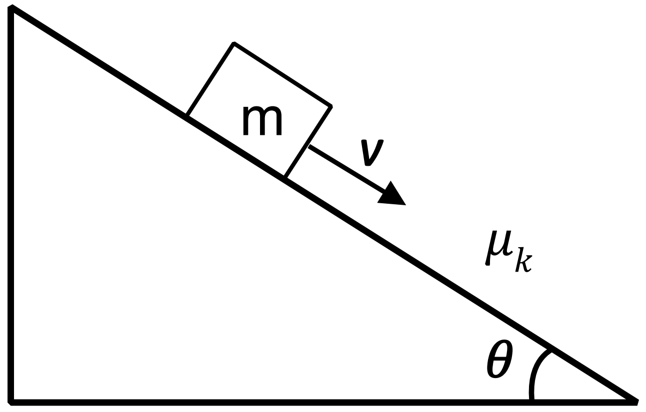
Case problem 3

A block of mass m moves down an inclined plane of angle θ with a constant velocity v as shown below. The coefficient of friction between the block and the inclined plane is given by μ_k.

What is the value of μ_k in terms of m, g, v, and θ? (link)



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Case problem 3 - Solution

The free body diagram of the block is given below. This block has three forces acting on it.

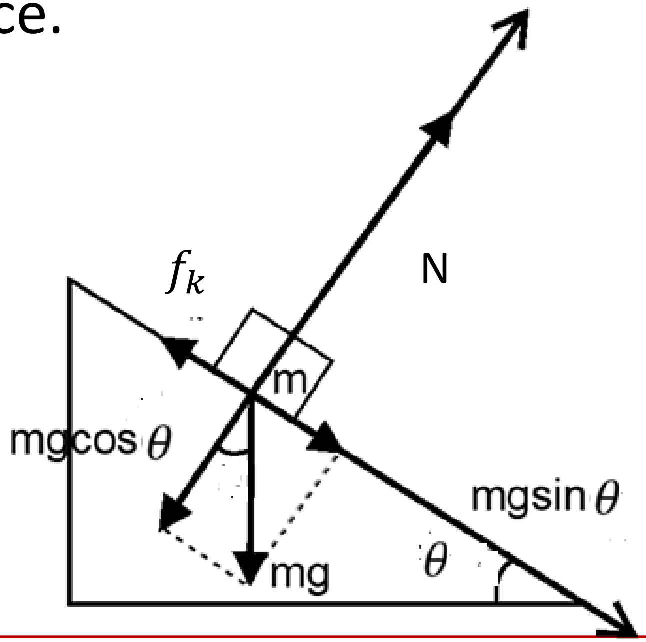
First, its weight under the influence of gravity, which is given as mg.

Second, the normal force of the plane, which is given as N.

Third, the friction force, which acts opposite to its direction of motion and is given as $\mu_k N$.

We choose a coordinate system so that our x-axis aligns with the motion of the block down the plane, and the y-axis aligns with the direction of the normal force.

Thus the friction force points in the negative direction of the x-axis, and the normal force is aligned with the positive direction of the y-axis. However, the weight mg is not along either of these axes, so we resolve the mg force into its components, $mgcos(\theta)$ along the negative y-axis, and $mgsin(\theta)$ along the positive x-axis.

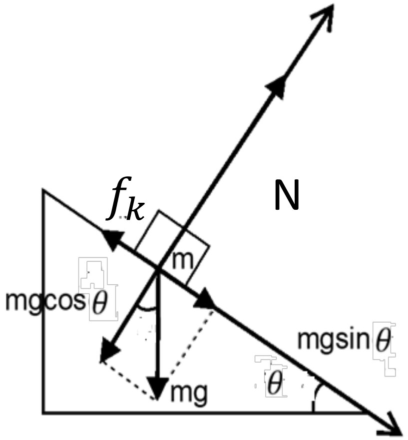


Case problem 3 - Solution

Newton's 2nd law gives us two equations:

$$\sum F_x = ma_x \longrightarrow \sum F_x = 0 \qquad \sum F_y = ma_y \longrightarrow \sum F_y = 0$$

Because the block is constrained to move along the surface of the inclined plane, there should be no acceleration in the y direction, and so $a_y = 0$. Also, because the block moves at constant velocity down the plane, Newton's 1st law assures us that there is no acceleration in the x direction as well, therefore $a_x = 0$. Plugging these accelerations in, we find that:



Summing all the forces in the x-direction gives us $\sum F_x = mgsin(\theta) - \mu_k N$

Summing all the forces in the y-direction gives us $\sum F_y = N - mgcos(\theta)$

Plugging these values into the force equations above gives us the following equations:

$$mgsin(\theta) - \mu_k N = 0$$

$$N - mgcos(\theta) = 0$$

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Case problem 3 - Solution

Solving for N in the second equation gives us $N = mgcos(\theta)$. Thus the normal force is equal to the cosine component of the weight. Substituting $mgcos(\theta)$ in for N in the first equation will give us the following:

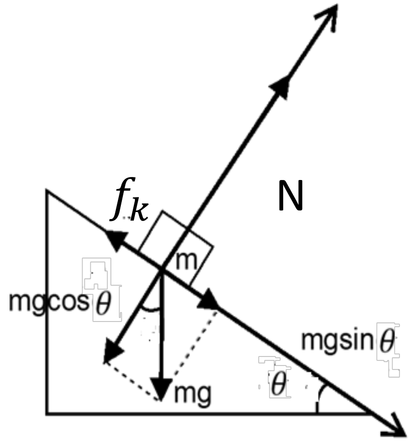
$$mgsin(\theta) - \mu_k mgcos(\theta) = 0$$

Now we solve the equation for μ_k. Adding $\mu_k mgcos(\theta)$ to each side gives us:

$$mgsin(\theta) = \mu_k mgcos(\theta)$$

Now we divide each side by $mgcos(\theta)$ to obtain:

$$\mu_k = \frac{mgsin(\theta)}{mgcos(\theta)} = tan(\theta)$$



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Off-class practice question(s)

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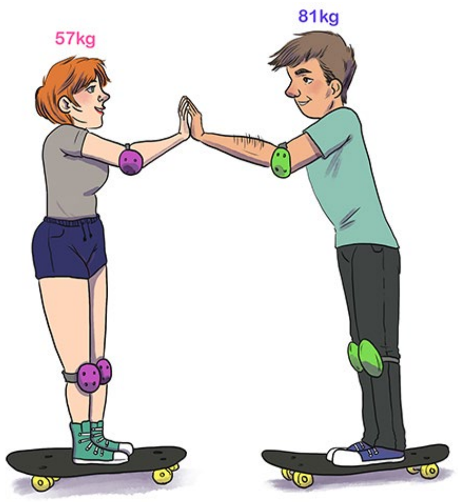
Case Problem

1. A baseball is thrown with an acceleration of $150\frac{m}{s^2}$ and its mass is $0.5kg$. What is the force? (link)
2. Two skateboarders stand facing each other.

   One skateboarder, the woman, weighing $57kg$ pushes the other skateboarder the man who is $81kg$ with a force of $85N$. What is the acceleration of each skater? (link)



https://mammothmemory.net/physics/newtons-laws-of-motion/newtons-third-law--examples/newtons-third-law-examples.html

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Case Problem: Solution

1. $F = ma = 0.5kg \times \frac{150m}{s^2} = \frac{75kgm}{s^2} = 75N.$

2. Using Newton's third law, the force on the man must be equal and opposite to the woman’s. Since $F = ma$ we have:

   woman: $a = \frac{F}{m} = \frac{-85N}{57kg} = -1.49m/s^2$

   man: $a = \frac{F}{m} = \frac{85N}{81kg} = 1.05m/s^2$

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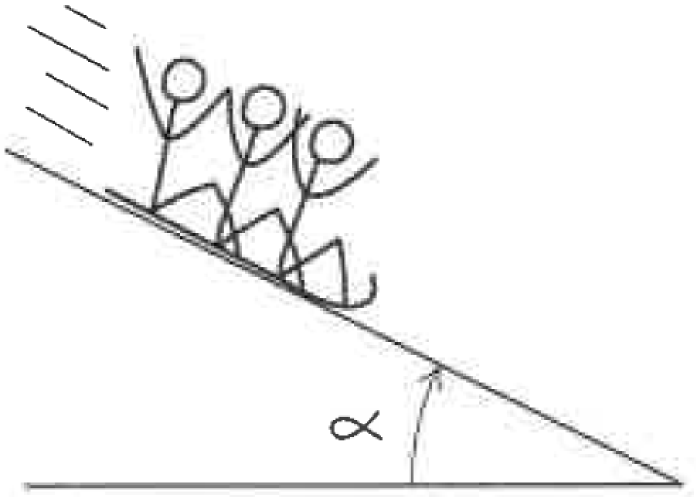
Case Problem

A toboggan loaded with students (total mass m) slides down a snow-covered hill that slops at a constant angle α. The toboggan is well waxed, so there is virtually no friction.

1. What is its acceleration?
2. What is the normal force on the toboggan by the hill?



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.10, p163

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Case Problem: Solution

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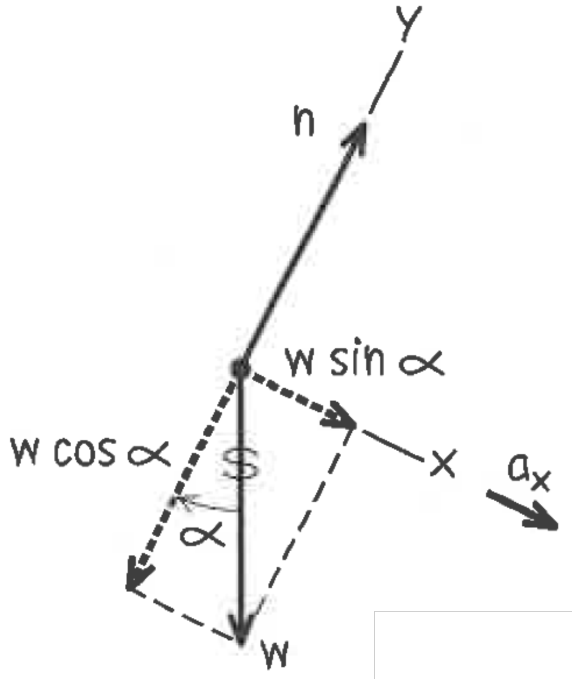

$$\sum F_x = ma_x$$

$$mg\sin\alpha = ma_x$$

$$a_x = g\sin\alpha$$

$$\sum F_y = ma_y$$

$$N - mg\cos\alpha = ma_y = 0$$

$$a_y = 0$$

$$N - mg\cos\alpha = 0$$

$$N = mg\cos\alpha$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.10, p163